



EXAM PAPERS PRACTICE

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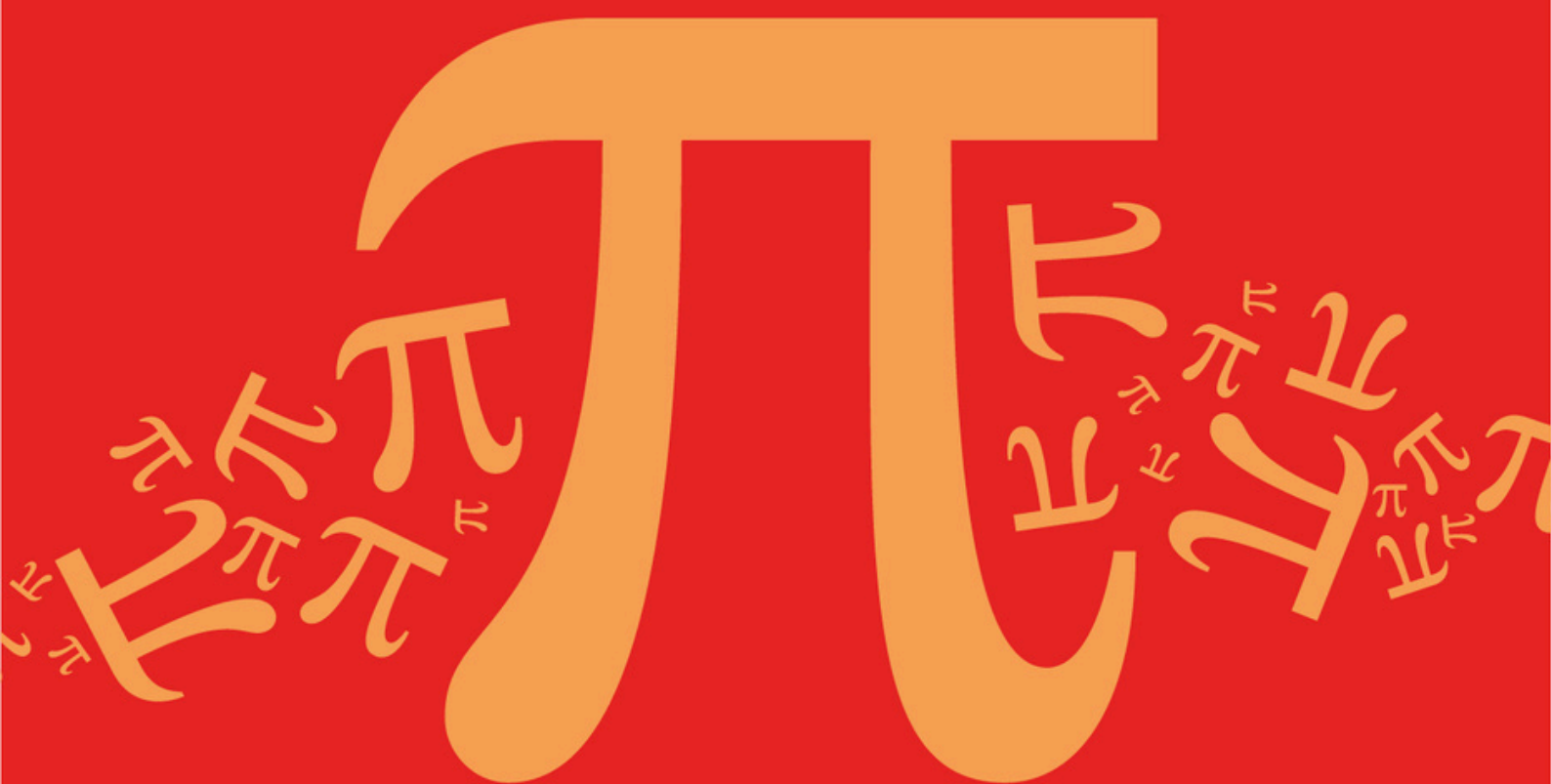
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DC: Network Flows

Sub topic

DC2: Understand Cuts

Mark Scheme

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Mark Scheme

Topic	DC: Network Flows
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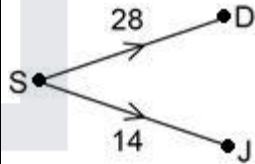
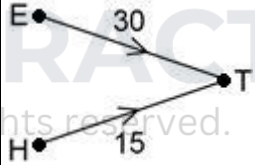
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

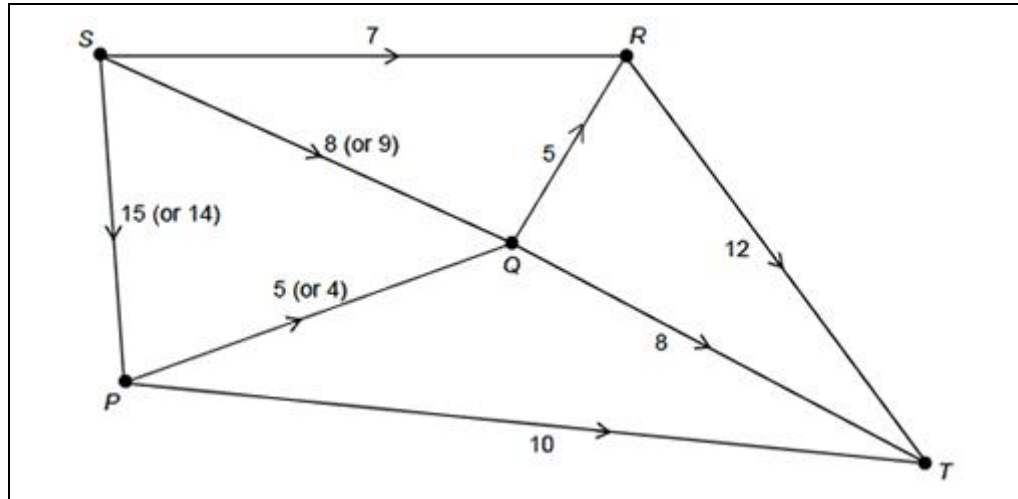
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the value of the cut	AO1.1b	B1	37
(i)				
(ii)	Deduces a weak inequality for the value of max flow based on their value of the cut Must see $\leq$ their value of the cut OE	AO2.2a	B1F	Max flow ≤ 37
(b)	Lists both sources CAO	AO1.1b	B1	D and J
(i)				
(ii)	Adds supersource S with directed arcs to D and J labelled with weights at least 28 and at least 14 respectively	AO1.1b	B1	
(c)	Lists both sinks CAO	AO1.1b	B1	E and H
(i)				
(ii)	Adds supersink T with directed arcs to E and H labelled with weights at least 30 and at least 15 respectively	AO1.1b	B1	
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Determines correctly the value of both cuts	AO1.1b	B1	35 and 36
(b)	Deduces the maximum flow in 'their' network by use of the maximum flow-minimum cut theorem and explains their reasoning.	AO2.4	E1	Max flow = 30 as max. flow = min. cut and the value of the minimum cut is 30



(c)	Determines correctly the flow along the saturated arcs	AO1.1b	B1	$SR = 7$, $QR = 5$, $RT = 12$, $QT = 8$, $PT = 10$
	Determines correctly a possible flow for each of SP , SQ and PQ	AO1.1b	B1	$SP = 15$ and $SQ = 8$ and $PQ = 5$ or $SP = 14$ and $SQ = 9$ and $PQ = 4$
Total 4 marks				

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains correctly using maximum flow into and minimum flow out of A	AO2.4	E1	Max flow into vertex A = 7 Min flow out of vertex A = $4 + 3 = 7$ As flow into A = flow out of A, both AD and AB must be at the lower capacity
(i)				
(ii)	Explains correctly using maximum flow into and minimum flow out of E	AO2.4	E1	$AD = 4$ from (a)(i), so $DE = 1$ and $DT = 3$. Since flow out of vertex E is at least 2, and $DE = 1$, BE must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut $\{S, A, B, C\} \setminus \{D, E, G, T\}$ $= AD + BE + \min(BG) + \min(CG)$ $= 4 + 1 + 5 + 2$

			$= 12 > 11$
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(c) (i)	Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with SA, SC, AB, BC and CF correct.	AO1.1a	M1	See diagram below												
	Determines correctly the potential increase and decrease for each arc with values on all arrows correct.	AO1.1b	A1													
(ii)	Finds correctly one augmenting path and flow	AO1.1b	B1	<table><tr><th>Augmenting Path</th><th>Flow</th></tr><tr><td>SADEFT</td><td>1</td></tr><tr><td>SCBET</td><td>3</td></tr><tr><td>SADBET</td><td>1</td></tr><tr><td> </td><td> </td></tr><tr><td> </td><td> </td></tr></table>	Augmenting Path	Flow	SADEFT	1	SCBET	3	SADBET	1				
	Augmenting Path	Flow														
	SADEFT	1														
	SCBET	3														
SADBET	1															
Finds correctly two augmenting paths and the flows	AO1.1b	B1														
Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units	AO3.2a	B1	Maximum flow through the network is 22 litres per second													
(iii)	Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of and clearly state that the value of this cut is equal to the value of the	AO2.1	R1	$\{S, A, B, C, D, E\} / \{F, T\} = 7 + 3 + 9 + 3 = 22$ As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.												

flow in (c)(ii) and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)			
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(iv)	Argues that, as 7 litres per second are initially flowing through node <i>E</i> , only a further 2 litres per second can flow through node <i>E</i>	AO2.4	R1	Flow through network can only increase by 2 litres per second as <i>DT</i> and <i>CF</i> are already saturated Therefore restricted capacity node reduces the maximum flow to $17 + 2 = 19$ litres per second
	Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1	
				Total 11 marks

Q4.

(a) *ABEH* 8

ACFH 5

ADGH 11

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B1

1

(b) (i) *ACEH* 2

One correct route and flow

M1

ACGH 4

At least one other correct

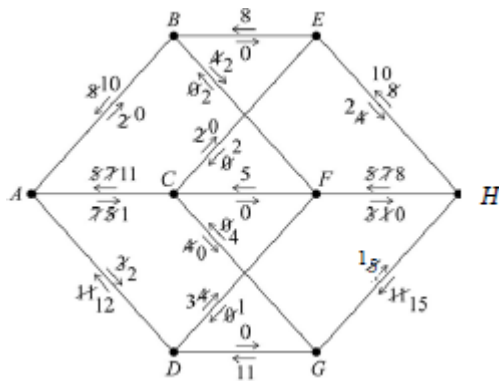
A1

Either *ADFH* 1 and *ABFH* 2

All correct

A1

Or *ADFH* 3



Forward and back flows on diagram

All correct

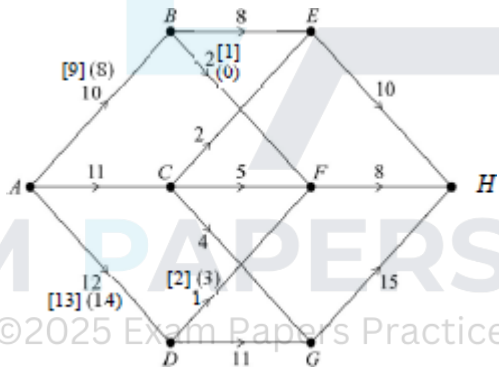
M1

A1

5

(ii) Max flow 33

B1



OE

B1

2

(c) Cut through BE, CE, FH, CG, DG

B1

1

[9]

Q5.

(a) 19

B1

1

(b) E

B1

1

(c) C

B1

1

(d) $x = 8$
 $y = 13$
 $z = 39$

B1 x 3

3

(e) 76

B1

1

(f) 83

B1

1

[8]

Q6.

(a) (i) $18 + (0 + 10 + 3 + 5) (= 36)$

B1

1

(ii) 30, 32, 36 (missing cut values)
B1 each value correct

B3

3

(iii) Max flow = 29
Award B0 E1 if their min^m cut is < 29

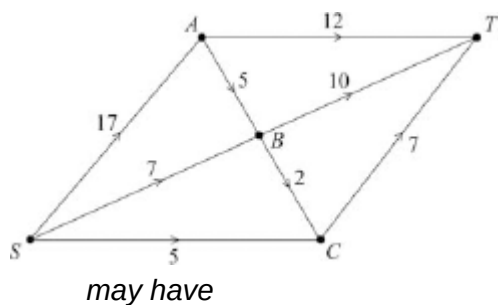
B1

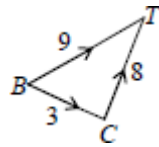
because value of minimum cut is 29
and min^m value explained as max flow

E1

2

(iv)





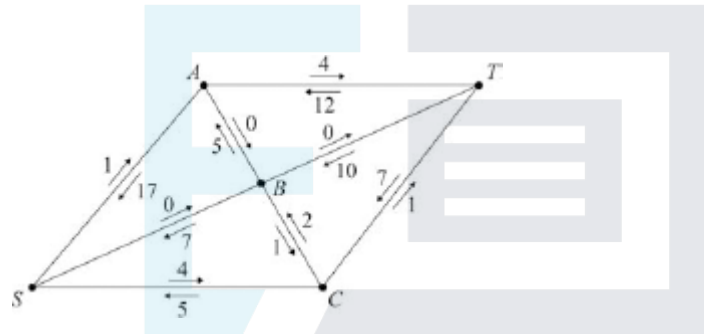
*see alternative
solution on next
page*

B1

cao

2

(b) (i)



*potential flow (forward and back)
4 pairs 'correct' including SC and AT
fit their (a)(iv) provided $0 < \text{flow} < 30$*

M1

*all pairs correct (condone missing 0s)
fit their (a)(iv) if correct flow < 29*

A1 ✓

one correct flow in table

m1

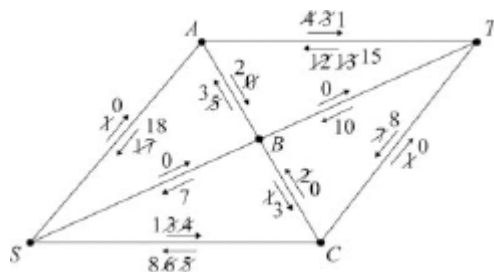


table
correct

SAT	1
SCT	1
SCBAT	2

*If (a)(iv) flow < 29 then may score A1 for correct table
giving max flow of 33*

A1

(see also the alternative solution)
 modifying flows (forward and back)
 1 flow correct ft their initial flow

m1

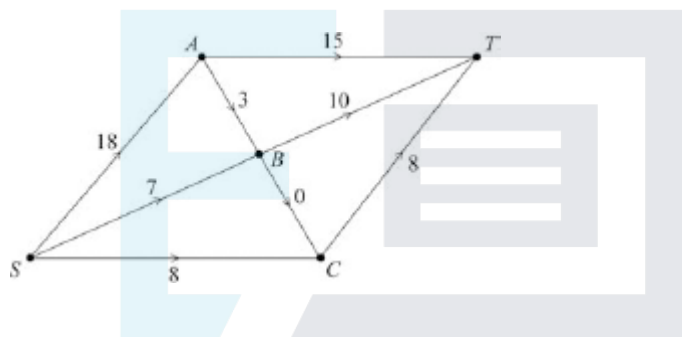
modified flows all correct, including all 0s (may score A1
 from a correct flow < 29 seen in (a)(iv) if final flow correct)

A1

6

(ii) new max flow = 33

B1



6 flows correctly interpreted from their labelling procedure
 provided M2 or M3 scored in (b)(i)
 (may have AB 2, AT 16, BT 9 – see over)

M1

flow correct

SC B1 if flow of 33 shown correctly but not from correct
 labelling procedure

A1

3

[16]

Q7.

(a) Cut value = $40 + 27 + 0 + 24$
 $= 91$

B1

1

(b) *ABDX* 16

B1

GFBX 18

B1

GHEX 20

B1

3

- (c) (i) One correct route with additional flow
any feasible route and flow

M1

Another 2 routes and flows correct
total flow at least 80

A1

All routes correct with total flow = 85

A1cso

Forward and backward flows on diagram
(directions must be clear)
at least 8 edges with pairs of values 'correct'

M1

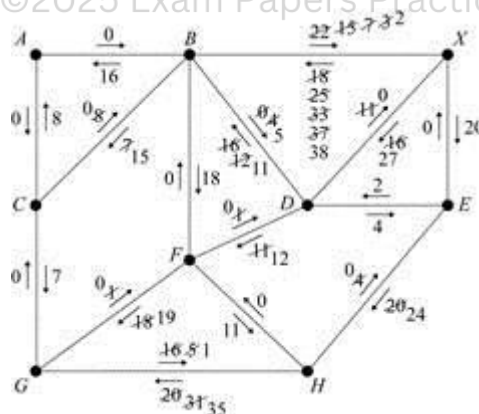
Augmenting flows
correct

A1cso

5

Consider other possible correct flows

Condone diagram as shown but really
should have initial flows in DE, etc



Route	Flow
ABDX	16
GFBX	18
GHEX	20
GCBX	7
ACBX	8
GHFDX	11
GHEDBX	4
GFDBX	1

- (ii) Max flow = 85

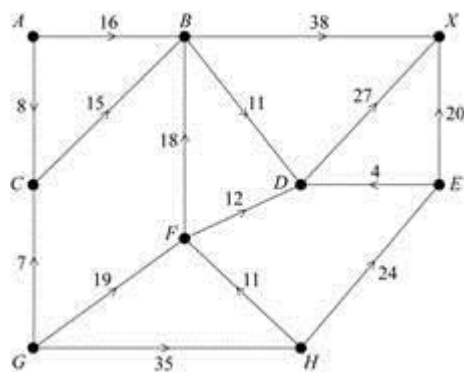
B1

Correct max flow

B1

2

Consider other possible correct flows



- (d) Considering 'their' $AB+CB+FB - 45 = 4$ fewer

M1

$\Rightarrow$ max number = 81

A1cao

2

[13]

Q8.

- (a) (i) Value of cut = $38 + 25 + 0 + 0 + 34$

Must show correct addition

= 97

AG

B1

1

- (ii) $\{S, A\}, \{B, C, T\}$ 65

B1

$\{S, B\}, \{A, C, T\}$ 57

B1

$\{S, B, C\}, \{A, T\}$ 72

B1

$\{S, A, B, C\}, \{T\}$ 56

B1

4

- (iii) Maximum flow = 53
ft their least cut value

B1ft

Minimum cut = Max flow

E1
2

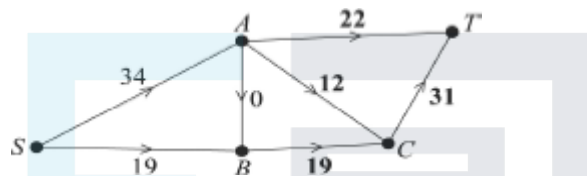
(iv) Their max flow on SA , SB or AT , CT

M1

All correct

A1
2

AT 22; AC 12; BC 19; CT 31
 $AB = x$; $AS = x + 34$; $SB = 19 - x$
 $0 \leq x \leq 4$



- (b) (i) Initial flow on Figure 3
Forward potential and backward flow
Condone 2 slips, ft their Figure 2



M1

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One correct augmented path in table and correct flow

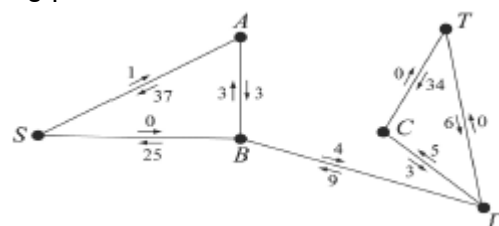
Path	Additional Flow
$SBDT$	6
$SABDCT$	3

M1

Table correct with total additional flow = 9

A1

Final network correct with evidence of
labelling procedure used

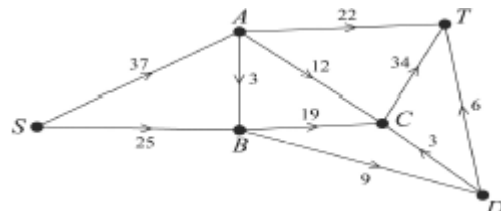


A1
4

- (ii) New maximum flow = 62

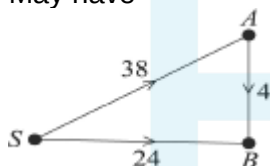
B1

Correct maximum flow on network



B1

May have



2

[15]

Q9.

- (a) Value of cut = $10 + 10 + 15 - 4 - 1$
condone one slip if working shown

M1

= 30

A1
2

- (b) BT 2, DE 3, ET 12
any 2 correct

B1

all correct

B1
2

- (c) (i) Initial flows forward and back or double
 Arc with arrows (at **least 6 pairs correct**)
*Condone pairs of values, (coordinates)
 with single arrow*

M1

*all correct (condone pairs with single
 arrow provided key indicated)*

A1

2

(ii)

Path	Flow
<i>SABT</i>	2
<i>SCDET</i>	1
<i>SACBT</i>	1

(or SCBT instead of SACBT with flow 1
first correct path and flow

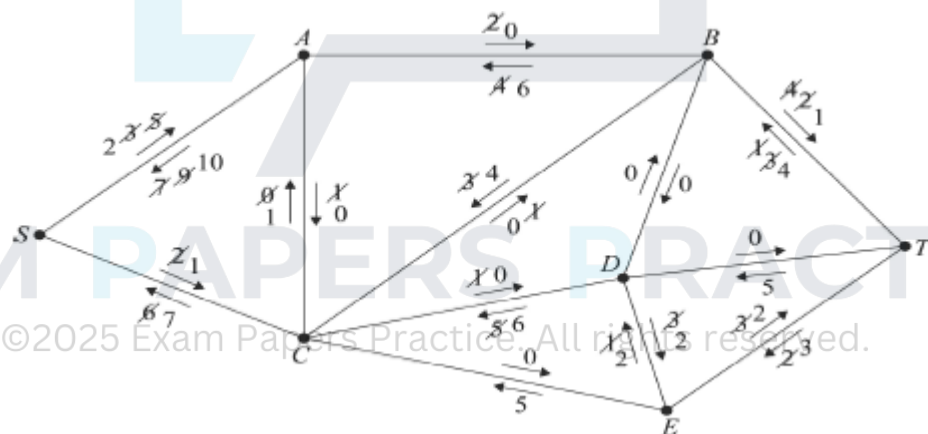
M1

another correct path and flow

A1

*all correct
 (other possibilities also)*

A1



Must have forward and backward flows
augmenting flows (6 pairs correct)

M1

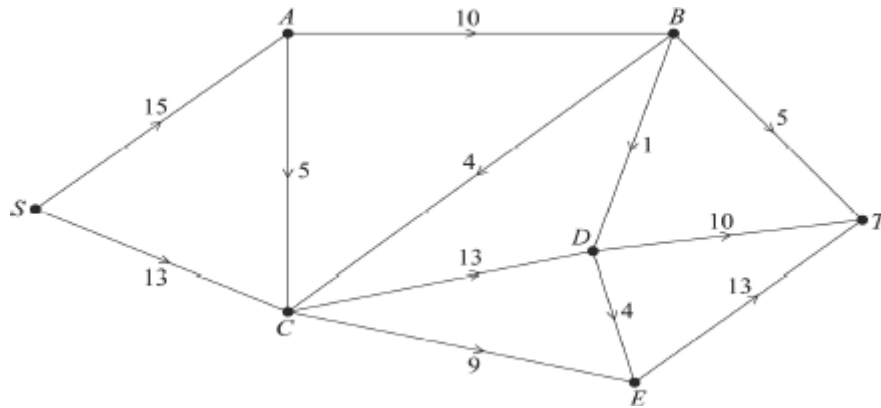
correct

A1

Alternative SA (3 & 9) SC (0&8)

5

(iii)



May have
 SA (14), SC (14)
 and AC (4) using
 alternative

Maximum flow values

*at least 8 correctly interpreted from their
 Figure 3 but $24 < \text{their maxflow} < 29$*

M1A1

2

- (d) Cut through AB, CB, CD and CE

May use $\{S, A, C\} / \{B, D, E, T\}$

But must have total flow of 28 in their
 network (condone one slip)

B1B1

1

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[14]

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Q10.

- (a) Arrival gates are U and R

B1

1

- (b) Cut value = $45 + 53 + 20 + 37 + 0$
 $= 155$

B1

1

- (c) Max flow along $UTSP$ is 17

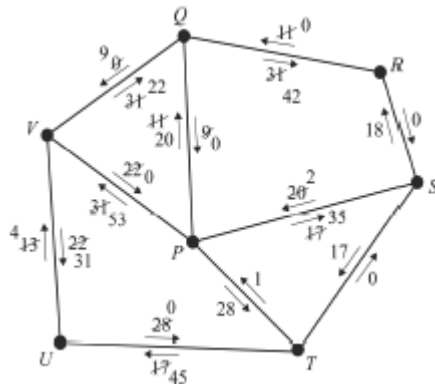
B1

and along $RQVP$ is 31

B1

2

- (d) (i)



Route	Value of Flow
<i>UTSP</i>	17
<i>RQVP</i>	31
<i>RSP</i>	18
<i>RQP</i>	11
<i>UTP</i>	28
<i>UVP</i>	22
<i>UVQP</i>	9

Initial flows along *UTSP* and *RQVP* with potential increases and decreases

B1

Table: first route and correct flow
After *UTSP* and *RQVP*

M1

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Another route and flow

A1

Table correct

A1

Network: attempt to use labelling procedure with forward/backward flows

M1

All diagram correct

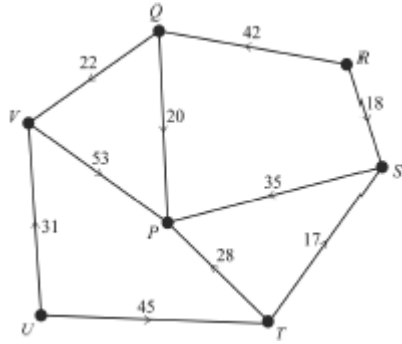
A1

6

(ii) Maximum flow = 136

B1

Figure 2 correct:



Other possible answers

B1
2

- (e) Rate reduced by 3
"their" maximum flow – 3

New maximum is 133

M1

A1
2

[14]

Q11.

- (a) Value of cut = $30 - 10 + 12 + 20$

= 52

Full marks for correct answers without working

A1
2

- (b) $AE = 9$;

$EF = 5$;

$FG = 4$

B1

B1

B1

3

- (c) (i) Attempt at forward and backward flows
 $SA \ 2 \ \& \ 4$; $AB \ 1 \ \& \ 3$; $BT \ 1 \ \& \ 3$

SD 3 & 1; DA 0 & 3; AE 0 & 3
 BE 0 & 7; DE 2 & 0; ET 1 & 3
 FD 2 & 1; EF 5 & 1; EG 1 & 5
 FG 1 & 2; GT 3 & 0

At least 5 pairs correct

10 pairs correct

all correct

M1

A1

A1

3

(ii) First flow augmenting path and correct flow on table

Table correct

Adjusting flows – forward and back

Correct

M1

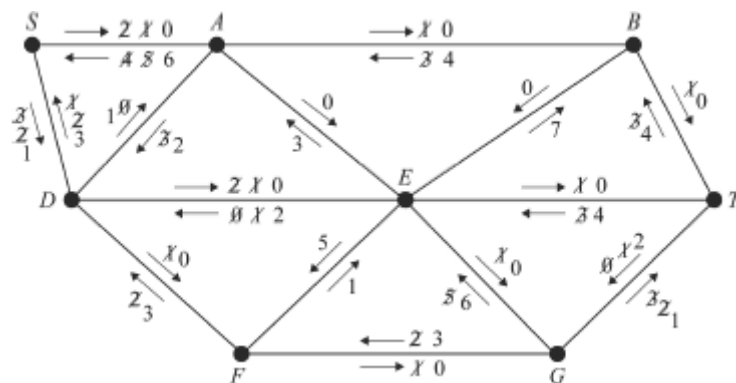
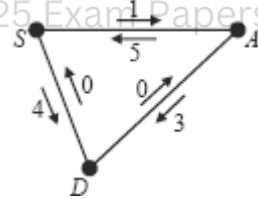
A1

M1

A1

4

May end up with



Path	Extra flow	Or $SDET$
$SABT$	1	
$SADET$	1	
$SDFGT$	1	

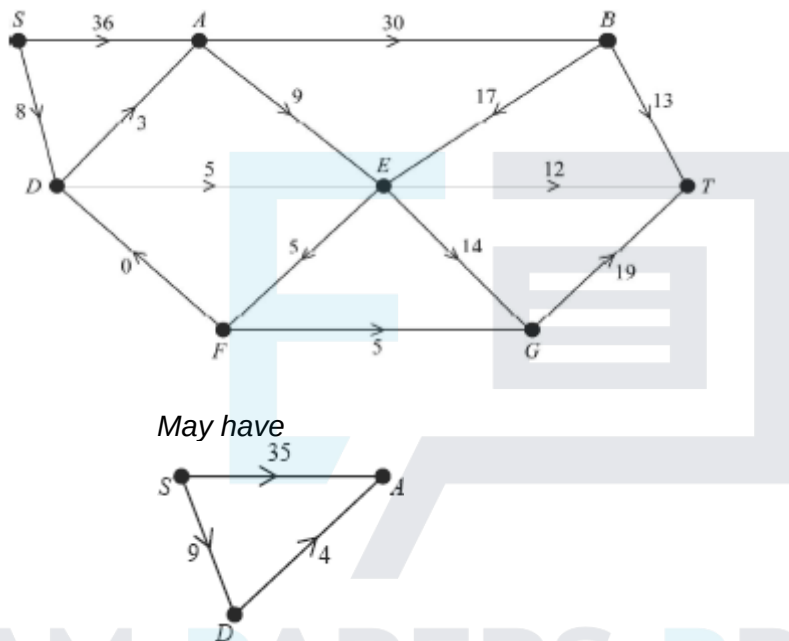
SDEGT 1 |

- (d) Max flow of 44 shown on figure 3
up to 2 slips

M1

all correct

A1



- (e) Cut through their saturated arcs

M1

Cut passes through *AB, AE, DE and DF*
Or *BT, ET, EG, and FG*

A1

2

[16]

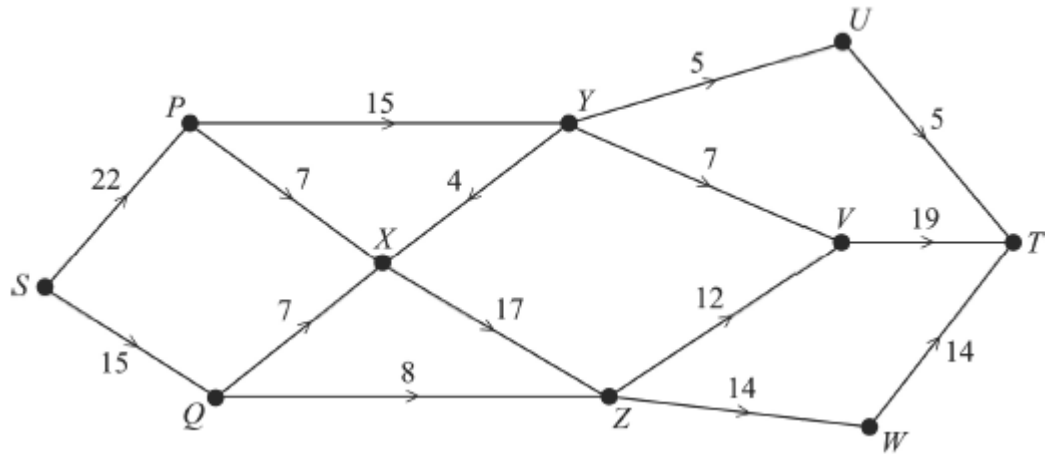
Q12.

- (a) Correct position of *S* and *T*

M1

Values on edges *SP, SQ, UT, VT* and *WT*

A1



2

- (b) (i) Cut C has value 40
 $15 + 0 + 17 + 8$

B1

1

- (ii) Max flow ≤ 40

E1

1

(c)

Route	Flow
$SQZWT$	8
$SPYXZVT$	4

B1B1

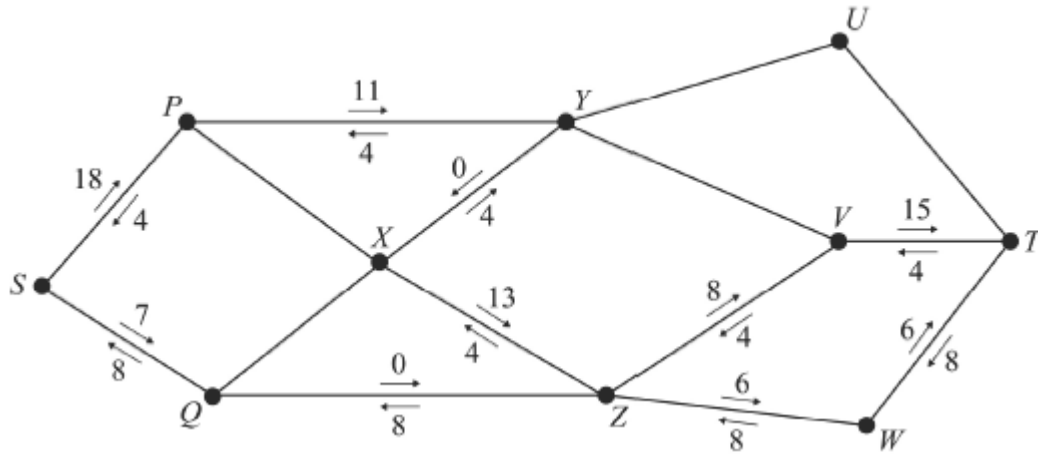
2

- (d) (i) 3 forward and backward flows correct

M1

All initial values correct on edges below

A1



2

(ii)

Route	Flow
SQZWT	8
SPYXZVT	4
SPYUT	5
SPYVT	6
SPXZVT	7
SQXZWT	6
SQXYVT	1

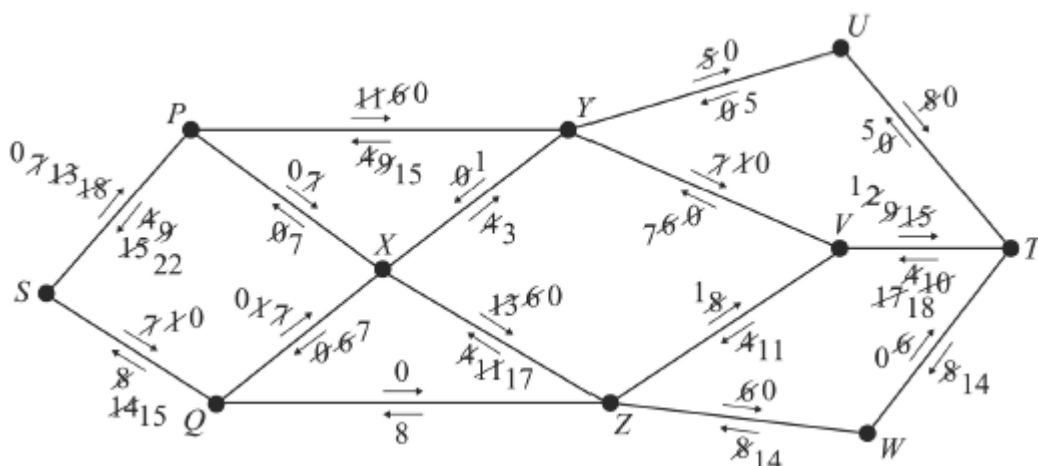
(Many different possibilities)
2 or more correct flows in table

Table correct (adding to 37)

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At least 2 flows augmented on diagram

Correct forward and backward final flows



4

Other possibility for ZV, VT, ZW and WT

- (e) Flow from Y to X is 3

B1

1

[13]

Q13.

- (a) (i) $17 - 9 + 16 + 20 = 44$

B1

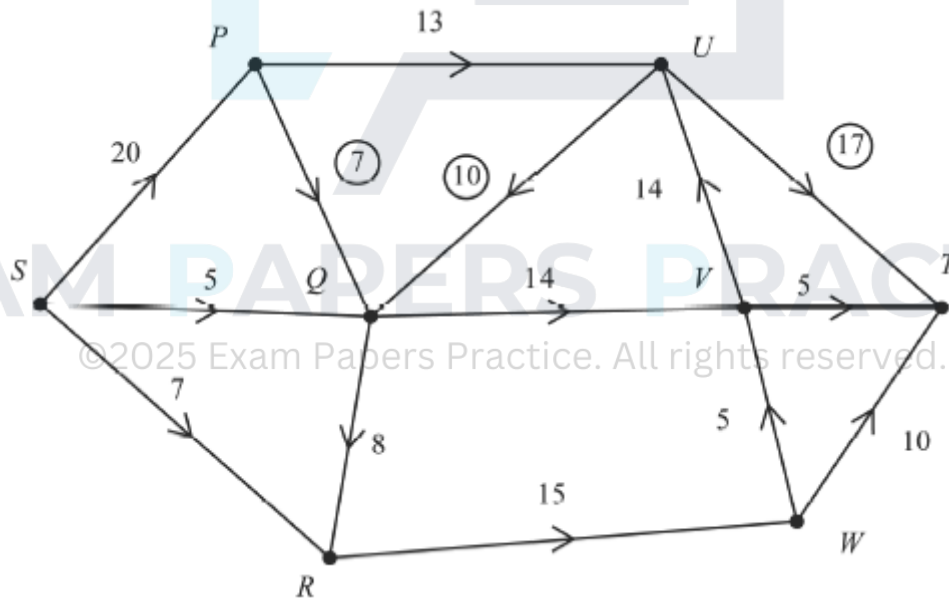
1

- (ii) Max flow ≤ 44

B1ft

1

- (b)



7

B1

10

B1

17

B1

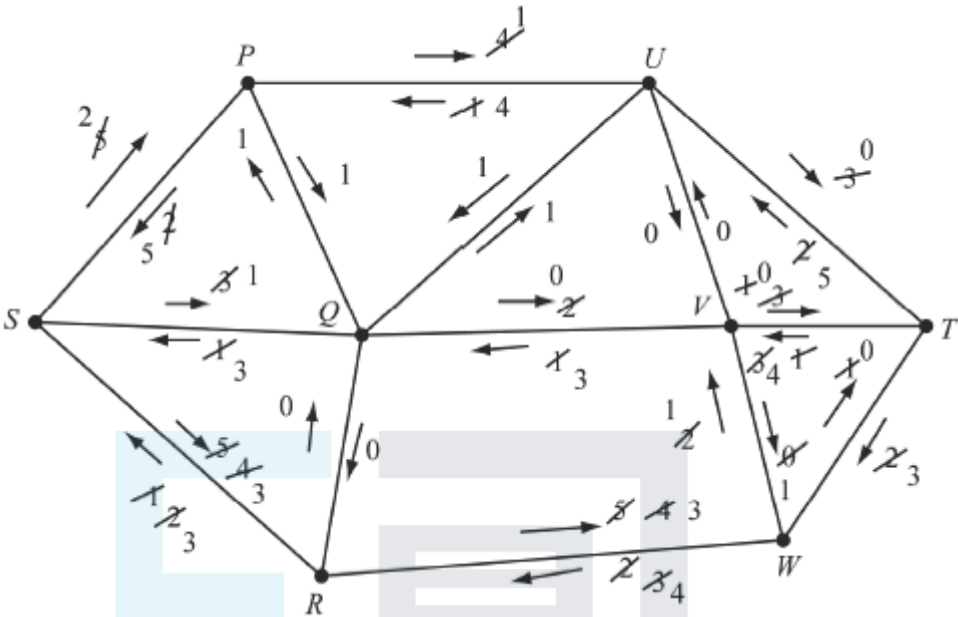
3

- (c) (i) Initial forward and backward flows
5 pairs correct

M1

Correct

A1



2

(ii)

Path	Additional Flow
SPUT	3
SQVT	2
SRWT	1
SRWVT	1

adjusting flows on network (1 path shown correctly)

M1

correct

A1

additional flow in table

M1

second flow

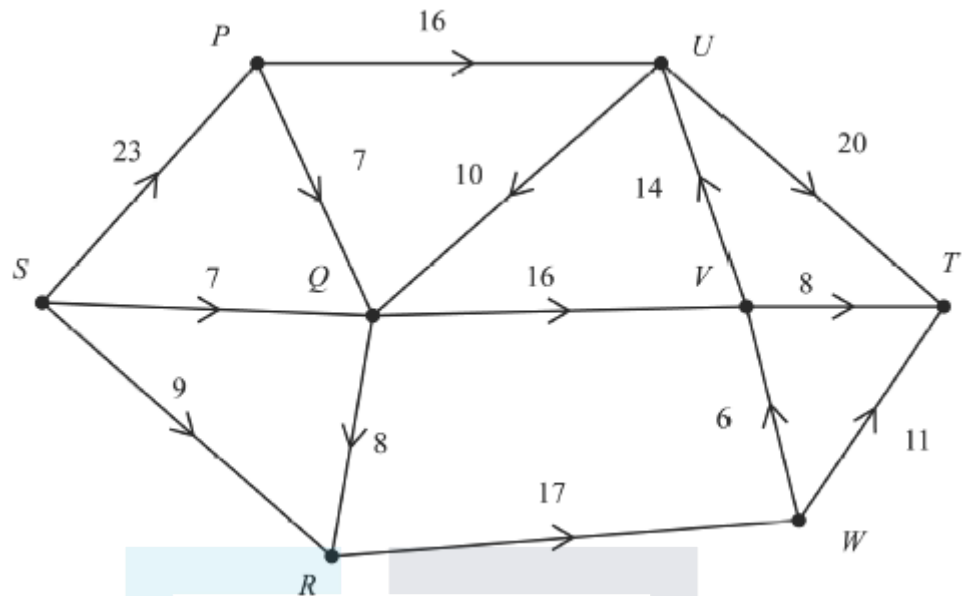
A1

all correct

A1

5

(iii)



Max flow of 39 (several possibilities of final flow diagram)

B1

1

[13]

Q14.

(a) (i) $15 + 0 + 14 + 7 + 9 = 45$

B1

1

- (ii) Maximum flow ≤ 45
 $\leq$ their value or < 45

M1

correct

A1

2

- (b) *SABT* flow 10
SDET flow 14
SFT flow 9
one correct

B1

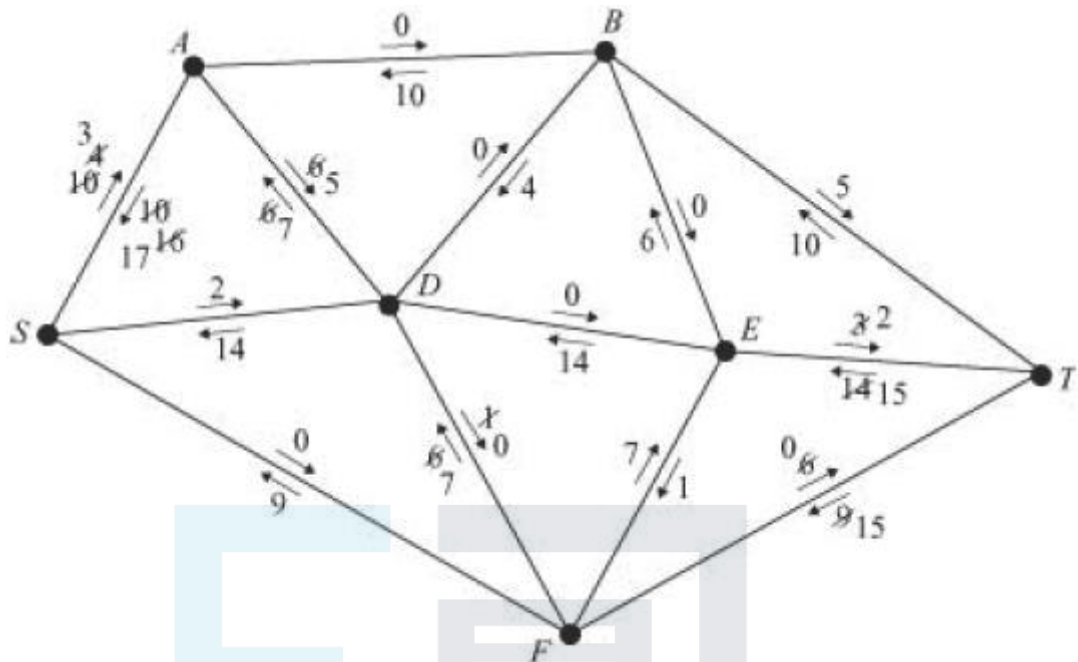
two more correct

B1

(may appear in table below)

2

(c) (i)



Additional route

M1

with correct flow

A1

one more correct route and flow

A1

table complete

correct total flow of 40

A1

correct use of potential and used flows

on network (may use double edges)

M1

values correctly updated

strict

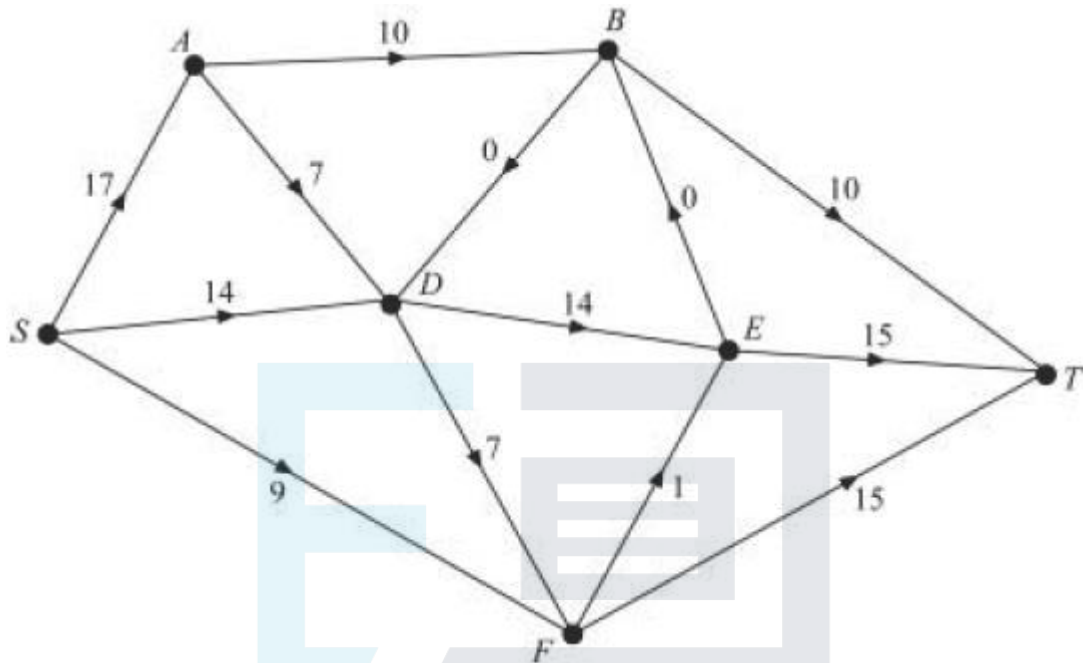
A1

6

Route	Flow
SABT	10
SDET	14
SFT	9
SADFT	6

SADFET	1
several possibilities	

(ii)



Maximum flow = 40

B1

Network showing flow of 40

B1

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(iii) Cut through saturated arcs
 AB, BD, DE, DF, SF

M1

Minimum cut shown to be 40 with statement
 linking to maximum flow

A1

2

[15]

Q15.

(a) (i) $5 + 8 + 16 - 3 = 26$

B1

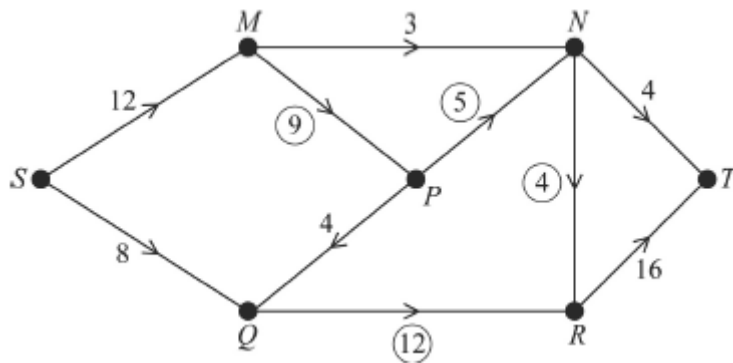
1

(ii) Max flow ≤ 26

E1ft

1

(b)



$MP - 9$

$PN - 5$

$NR - 4$

$QR - 12$

B1

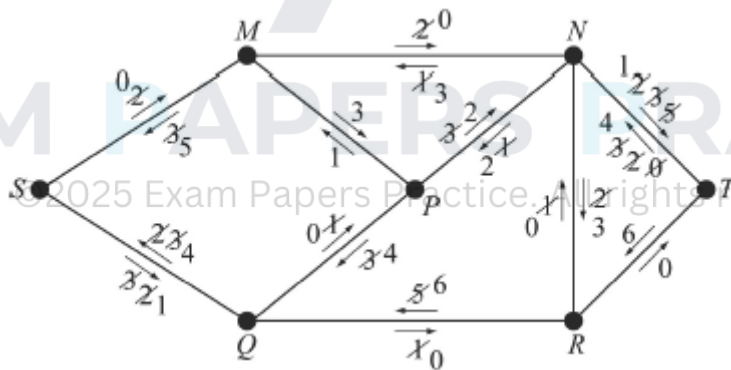
B1

B1

B1

4

(c) (i)



*initial flow – forward and backward
6 pairs correct*

M1

correct

$OM\ 2\ \&\ 3;$ $MN\ 2\ \&\ 1$

$NT\ 5\ \&\ 0;$ $MP\ 3\ \&\ 1$

$SQ\ 3\ \&\ 2;$ $PQ\ 3\ \&\ 1$

$PN\ 3\ \&\ 1;$ $QR\ 1\ \&\ 5$

$NR\ 2\ \&\ 1;$ $RT\ 0\ \&\ 6$

A1

2

(ii) Adjusting flows on network

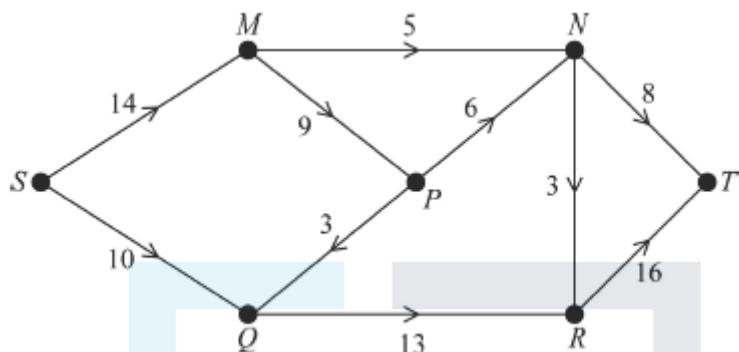
M1A1

Path	Flow
SMNT	2
SQPNT	1
SQRNT	1

First correct path and flow B1
 Second correct B1
 Rest B1

5

(iii)

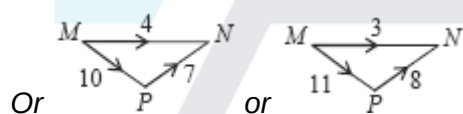


6 flows correct

all correct

M1

A1



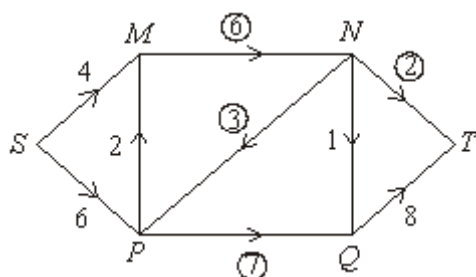
2

EXAM PAPERS PRACTICE [15]

Q16.

(a)

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MN

B1

NT

B1

PQ

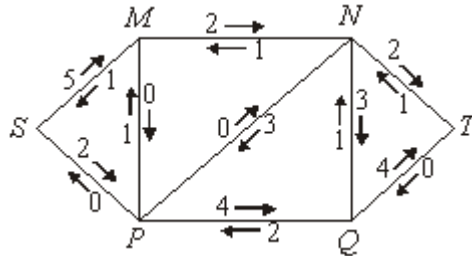
B1

NP

B1

4

(b) (i)



initial flow indicated as surplus forward and backward flows

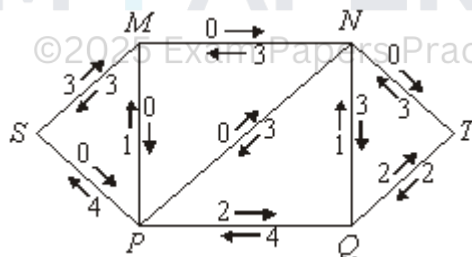
use of flow augmentation

e.g. SMNT 2

one flow correctly identified

SPQT 2

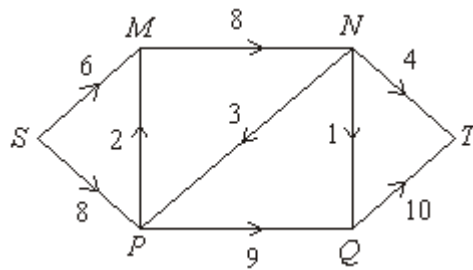
all possible flows correct



amending flows (dep on first M1)

final situation with saturation at M and P

(ii) Max flow – 14



B1 2

- (c) Cut through 2 of their saturated arcs
cut on **original** network

described or drawn

M1
A1 2

[14]

Q17.

- (a) D

B1 1

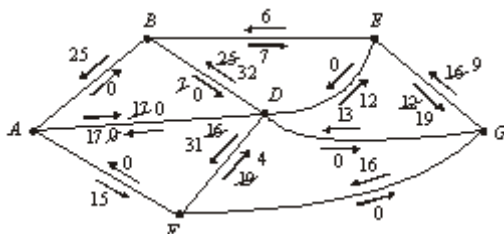
- (b) $(17 + 25 + 35 + 13 + 12 + 13 = 115)$

B1 1

- (c) $ABD_{\max} = 25$; $GED_{\max} = 12$

B1B1 2

- (d) (i)



Forward and backward flows

M1

Adjusting flows on diagram

M1

Routes and flows in chart

M1

One correct other than ABD, GED

A1

Another correct

A1

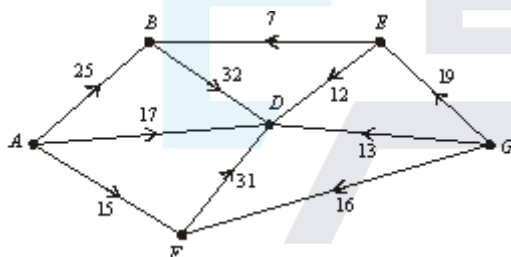
Route	ABD	GED	GFD	GD	AD	AFD	GEDB
Flow	25	12	16	13	17	15	7

All correct

A1

6

- (ii) Total = 105
Max flow



B1

B1

2

- (iii) Cut through AF, AD, BD, DE, DG, and GF
Through 3 saturated arcs (fairly generous)

M1

Correct

A1

2

- (e) Reduce max flow by their EG
changing 19 to 15
Reduce by 4 since everywhere else saturated

M1

⇒ New max = 101

Correct answer ⇒ 2 marks

A1

2

[16]

Q18.

(a) (i) $9 + 7 + 0 + 9 + 13 = 38$

B1

1

(ii) Maximum flow is less than or equal to 38
< their value of cut

M1

≤ 38
M0 for "equals" their cut

A1ft

2

(b) *SUYWT* flow of 9

B1

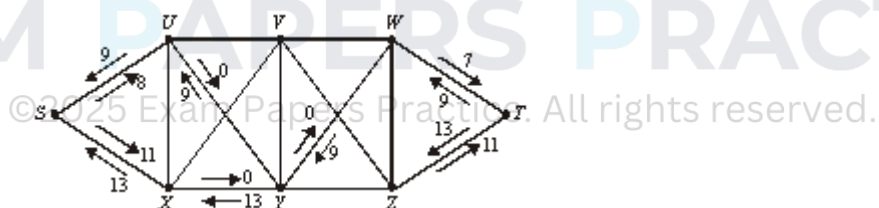
SXYZT flow of 13

B1

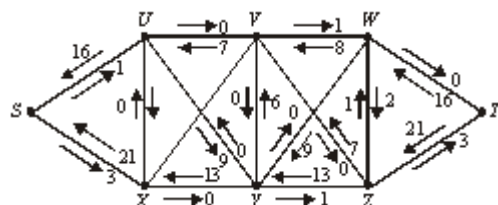
2

(c) (i) Indicating flows from (b) on network
Preferably as backward flows

M1



with augmentation leading to something such as



Route	Flow
<i>SUYWT</i>	9
<i>SXYZT</i>	13
<i>SUVWT</i>	7

SXVZT	7
SXVWZT	1

Flow augmentation (many possibilities)

$SUVWT \quad 7$

M1A1

$SXVZT \quad 7$

m1A1

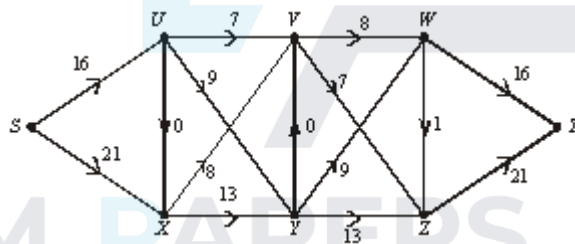
$SXVWZT \quad 1$

A1

6

- (ii) Network showing maximum flow
Several possibilities

B1



Maximum flow is 37
Or $\{S, U, X / V, W, Y, Z, T\}$

B1

2

- (iii) Attempt to find cut through saturated arcs

M1

Cut through UV, UY, XV, XY

A1

2

[15]

Q19.

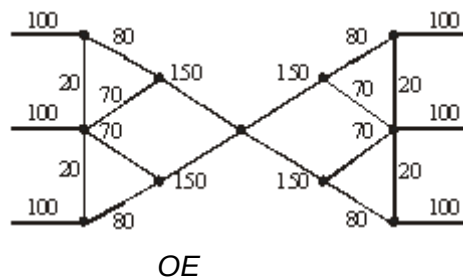
(a) $C_1 = 100 + 80 + 150$
 $= 330$

B1

$C_2 = 80 + 80 + 100 + 80 + 80$
 $= 420$

B1
2

(b)



M1A1A1
3

(c) (i) At I max = 300
 $\therefore$ restriction $\Rightarrow$ max = 120

B1
1

(ii) At H max = 150

M1

$\therefore$ restriction $\Rightarrow$ max = 270

A1
2

EXAM PAPERS PRACTICE [8]

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Q20.

(a) A, B, C, D

B1
1

(b) I

B1
1

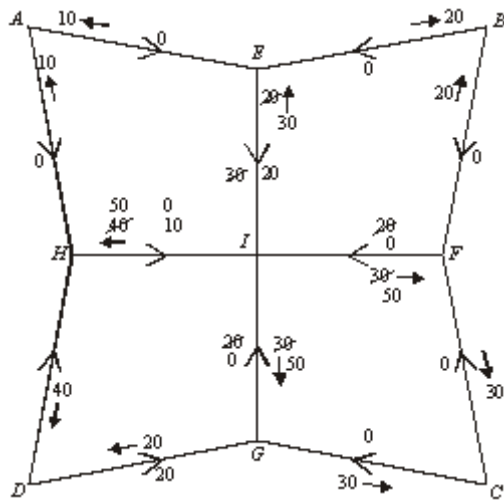
(c) $20 + 20 + 10 + 10 + 40 + 50 + 30$

M1

$= 180$

A1
2

(d)



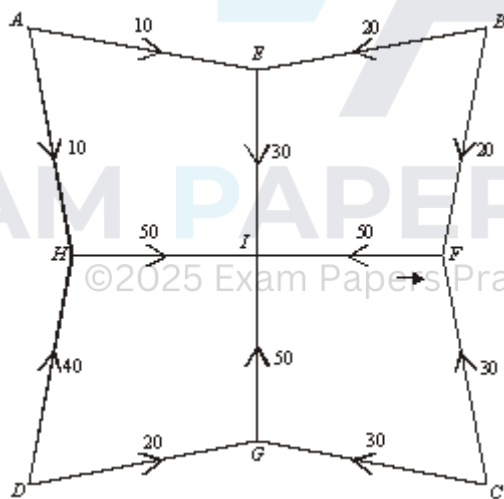
SCA

1 complete flow

M1

M1

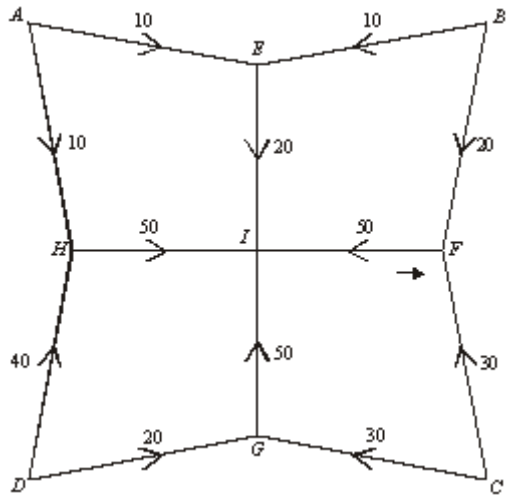
Final:



OE

A1

(e)



Their max – 10

Max flow = 170
CAO

M1

A1

2

[11]

Q21.

(a) $C_1 = 42$

B1

$$C_2 = 10 + 8 + 14 + 6 + 10$$

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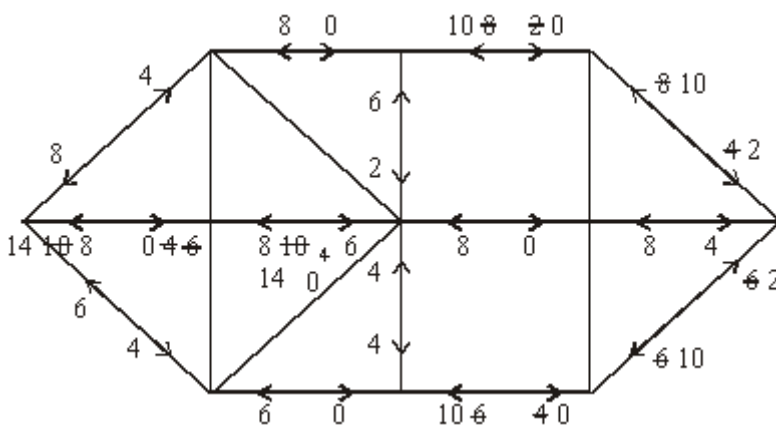
M1

$$= 48$$

A1

3

(b)



SCA

1 flow

M1*

2nd flow

M1*

1st

M1*

2nd

A1*

all correct
(*)OE

A1F*

Max flow = 28

A1*

B1

7

[10]

Q22.

(a) $C_1 = 40$

B1

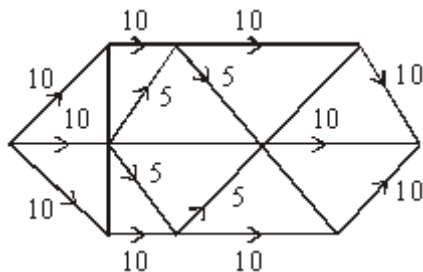
$C_2 = 35$

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B1

2

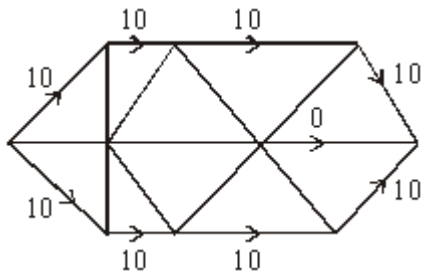
(b)



M1 A1

2

(c)



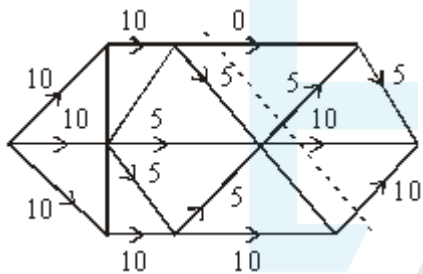
M1

Max flow = 20

A1

2

(d) (i)



M1

Max flow = 25

A1

2

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(ii) Min cut = 25

Attempted cut

M1

Min cut = Max flow

Correct cut GH, GJ, IJ or $(C_2 - EH)$

A1

2

[10]

Q23.

(a) (i) 30

(ii) 25

B1 B1

2

(b) $YZ = 5$ $XZ = 0$

M1 A1

$$WX = 5 \quad SW = 15$$

A1

3

(c) *SYZT*

M1 A1

SYWXT

M1 A1

4

(d) Total flow in (c) = $15 + 5 + 5 = 25$

Max flow $\leq$ any cut, so no flow can exceed the 25 in (a)

M1

(ii)

A1

2

[11]

Examiner reports

Q1.

The majority of students determined the value of the cut correctly, with the most common errors being numerical slips. Only 51% of students correctly interpreted their value of the cut, with many stating generic comments about the maximum-flow minimum-cut theorem but not making the connection between the value of any cut and the implications for the maximum flow.

Most students spotted the sources and sinks of the network, the most common error being to include a third node which was not a source or sink. However, less than half of the students correctly added arcs to the diagram, with common mistakes being the lack of arrows or correct capacities on each arc.

Q4.

Most students answered part (a) correctly. Although there were many correct solutions for part (b)(i), a significant number of students failed to give a convincing solution on their diagram. It is essential that students use 'backward' arrows to show existing flows and 'forward' arrows to show potential flows. There was a significant number of students who scored 5 marks in part (b)(i) but failed to score both marks in part (b)(ii). Part (c) was answered poorly. Students were asked to list the edges of their cut; many simply drew a cut on their diagram.

Q5.

Although many students scored the majority of marks on this question, full marks were rarely seen. Errors were common in answers to parts (e) and (f).

Q6.

- (a) (i) Many students were able to show the correct working to give the value of the cut.
- (ii) Far fewer were successful in finding the three values of the remaining cuts in the table.
- (iii) Some thought that the "maximum" cut was needed to give the maximum flow and others wrote " $\text{maximum flow} \leq \text{minimum cut}$ ". The 29 value had been given to help students and this proved valuable for those who had been unable to find the 3 correct values in the table.
- (iv) Those with a maximum flow of 29 were usually able to draw correct flows along the given edges, although some gave at least one value greater than the capacities on the original network.
- (b) Some missed the vital point of supplementing the flows by 4 along SC and AT and made no further progress. It was good to see many students trying to set out their solution in a logical manner and once again the diagram and table clearly helped. There are still many students unfamiliar with the "labelling procedure" who fail to show both potential forward **and** backward flows on their network. Students are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit students for their original

values if they have been obliterated during augmentation.

Some credit was given to students who had used flow augmentation correctly but had failed to find a maximum flow equal to 33, if they interpreted their flows correctly on the final diagram.

Q7.

- (a) Many candidates failed to obtain the correct value of the cut, usually as a result of subtracting 6 instead of adding 0.
- (b) Again, few had trouble finding the maximum flows along the given edges.
- (c) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation. This was a good discriminator and tested that candidates understood the labelling procedure. Only the best candidates realised the need to augment a flow in the direction from D to B ; not many obtained a flow greater than 80 and very few achieved the correct flow of 85 with a fully augmented correct diagram.
- (d) Most candidates realised that they needed to reduce their total flow by 4 because of the vertex restriction, but only the best candidates obtained the correct maximum number that could move through the network during the fire drill.

Q8.

Usually after crossing out one or two additional figures, in part (a)(i), most candidates managed to show that the cut had value 97. This(ii) had been intended as a help to find the values of the other cuts in part (a)(ii), but only the best candidates managed to find the 4 correct values in the table. In part (a)(iii), many candidates made statements involving inequalities instead of saying that the maximum flow was equal to the minimum cut. The cut with value 53 had already been provided in the table but several candidates ignored this value.

In part (b)(i), it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network; however, some candidates started from a flow of zero rather than using their values on Figure 2. Candidates are advised to use the table to show what additional flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. In part (b)(ii), those who had the correct maximum flow and who had used the labelling procedure correctly usually had no trouble in showing this maximum flow on the final diagram.

Q9.

Part (a) Most candidates seemed unaware of how to calculate the value of the cut correctly because of the upper and lower capacities. Some subtracted 9 instead of 5 from the sum of the other upper capacities; others merely added a set of positive numbers.

Part (b) In contrast, almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Part (c)(i) Some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge; one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can be shown in pencil so as not to obliterate the initial flow figures.

Part (c)(ii) A table was provided so that a flow of 2 along *SABT*, for example, could be listed in the table. The potential forward and backward flows along *SA*, *AB* and *BT* could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part(i) then they risk losing marks for that part. The majority of candidates found only 2 augmenting paths and were not able to find the correct maximum flow. Those who were not using the labelling procedure correctly sometimes obtained a flow that was impossible; for instance a maximum flow of 29 was seen quite often. It should be noted that candidates are not expected to do flow augmentation with negative flows.

Part (c)(iii) Some credit was given for transferring values from Figure 1 to Figure 2 even if the maximum flow had not been achieved; but full marks could only be scored for a correct flow of 28 through the network.

Part (d) Very few candidates realised the need to consider their saturated arcs in order to obtain the correct cut. Many wrongly considered their flow on Figure 2, where obviously every cut has a value of 28, and consequently wrote down cuts such as the one through *BT*, *DT* and *ET* as their final answer. That is why the question suggested that candidates “may wish to show the cut on the network above” since that network contained all upper and lower capacities.

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In part (a) most candidates identified the arrival gates and scored an easy mark.

In part (b) quite a few candidates made errors in calculating the value of the cut, with common wrong answers being 138 and 172.

In part (c) the values of the initial maximum flows were usually correct.

In part (d) it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made.

For part (d)(ii) the incorrect answer of 127 was seen far more often than the correct value for the maximum flow, since candidates seemed unaware of the need to introduce an additional flow of 9 by actually reducing the flow along some edges that were previously at full capacity.

In part (e) those who had the correct maximum flow in part (d) were usually successful in finding a new maximum flow and those who did not have a correct answer were given some credit for reducing their previous flow by 3 or explaining how the extra 3 passengers could be accommodated in some way.

Q11.

In part (a), most candidates seemed unaware of how to calculate the value of the cut correctly. Many subtracted 17 instead of 10 from the sum of the other upper capacities.

In contrast, in part (b), almost everyone scored full marks for finding the value of each of the missing flows along the given edges.

Some candidates still seemed unaware, in part (c)(i), of how to represent the potential increases and decreases **from the initial feasible flow**. It requires forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows so that any adjustments can best be shown in pencil so as not to obliterate the initial flow figures. In part (c)(ii), a table was provided so that a flow of 1 along $SABT$, for example, could be listed in the table. The potential forward and backward flows along SA , AB and BT could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. If candidates obliterate their values of the potential increases and decreases from part (c)(i), then they risk losing marks for that part. The majority of candidates found only 3 augmenting paths and were not able to find the correct maximum flow.

In part (d), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 44 litres per second.

In part (e), very few candidates realised the need to consider their saturated arcs in order to obtain a minimum cut. Many wrongly considered their flow on Figure 3, where obviously every cut has a value of 44, and consequently wrote down cuts such as BT , ET and GT as their final answer.

Q12.

Part (a) All candidates seemed familiar with the idea of a super-source, S , and super-sink, T . Although it was possible to assign equal values such as 100 to the capacity on each of the new edges, such a large value often caused problems with flow augmentation. Some incorrectly assigned equal values such as 20 to the new edges. Those using capacities of 22 on SP and 15 on SQ were usually the most successful.

Part (b) Quite a few candidates made errors in calculating the value of the cut with a common wrong answer being 36. Several incorrectly thought that the maximum flow was equal to the value of this cut.

Part (c) The values of the initial maximum flows along the given edges were usually correct.

Part (d) It was good to see candidates trying to set out their solution in a logical manner with most candidates showing potential forward and backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values, particularly the initial flow, should be clear to the examiner when such modification is made. It is not wise to totally obliterate previous values when adjustments are made.

Part (e) Some candidates misread this last part and stated the maximum flow through the network. Very few were able to use their final maximum flow diagram to interpret the correct flow from Y to X .

Q13.

In part (a), about half the candidates seemed unaware of how to calculate the value of the cut correctly. In part (b), by contrast, almost everyone scored full marks for finding the value of the missing flows along the given edges.

In part (c)(i), some candidates still seemed unaware of how to represent the potential increases and decreases **from the initial feasible flow**. It required forward and backward arrows or a duplicate edge: one showing potential forward flow, the other the potential backward flow. The initial flow values are best written in black ink close to the arrows, with any adjustments then shown in pencil so as not to obliterate the initial flow figures.

A table was provided in part (c)(ii) so that a flow of 3 along SPUT, for example, could be listed in the table. The potential forward and backward flows along SP, PU and UT could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. There are many correct solutions to this part, and teachers might find it informative to consider different possibilities with their future students in order to illustrate that, whatever solution they obtain, the maximum flow is still the same.

In part (c)(iii), those who used flow augmentation correctly usually had no trouble in completing the diagram to show a maximum flow of 39 litres per second.

Q14.

In part (a), a number of candidates made errors in calculating the value of the cut; a common wrong answer was 39. Several incorrectly thought that the maximum flow was equal to the value of this cut. In part (b), the values of the initial maximum flows were usually correct, and in part (c), it was pleasing to see candidates trying to set out their solution in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made and the final backward flows should be the values transferred onto Figure 1 to give a possible maximum flow. Although quite a few were successful in finding a cut of the same value as the maximum flow, they failed to convince examiners that they had proved that the flow must be a maximum.

Q15.

In part (a), very few candidates seemed prepared for a question with lower capacities as well as upper capacities, and consequently the correct value of the cut was rarely seen.

In contrast, in part (b), almost everyone scored full marks for the missing flows along the given edges.

In part (c)(i), many candidates seemed unaware of how to represent the potential increases and decreases from the initial flow. It requires forward and backward arrows or a duplicate edge: one showing the potential forward flow, the other the potential backward flow. In part (c)(ii), flow augmentation was not widely understood. A table was provided so that a flow of 2 along SMNT could be listed in the table. The potential forward and backward flows along SM, MN and NT could then be adjusted on the diagram by lightly crossing out the original flows along each edge and indicating the new values. Candidates who used flow augmentation correctly usually had no trouble in completing the diagram in part (c)(iii) to show a maximum flow of 24.

Q16.

The insert seemed to help candidates to set out their solution in a logical manner. Most

candidates completed the first network indicating the feasible flow. When doing the flow augmentation in part (b)(i), candidates rarely showed their working properly. It is in cases like this, with minimum and maximum capacities, that the formal method (i.e. two arrows and numbers for each edge showing potential increases and decreases of capacity) can be very helpful for candidates and they should be encouraged to always use it. Teachers might again find it valuable to consult the solution in the mark scheme and to encourage their students to take this approach. Also, in trying to find a cut of 14, only a minority of candidates used the original capacities in the network printed in the question paper.

Q17.

Part (a) was usually correctly answered but a number made errors in calculating the value of the cut in part (b). The values of the initial maximum flows in part (c) were usually correct also. The insert **Figure 4** was intended to help candidates to set out their solution to part (d) in a logical manner. Some candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. It should be clear to the examiner what the previous values were when such modification is made and the final backward flows should be the values transferred onto **Figure 5** to give a possible maximum flow. Very few were successful in finding a cut of the same value as the maximum flow. The final part (e) was well answered by most of those who had found the correct maximum flow.

Q18.

Part (a) A very common wrong answer for the value of the cut in part (i) was 44. This arose from assuming that the edge VY contributed 6 rather than zero to the value of the cut. A number of candidates wrote in part (ii) that the maximum flow was equal to the value of their cut and this response scored no marks.

Part (b) It was pleasing to see almost all the candidates finding the correct maximum possible flows along the two routes.

Part (c) The labelling procedure involves showing the flow already in the network, usually as a backward flow, together with the surplus potential flow as two distinct figures on the diagram. Candidates would benefit from seeing how the mark scheme indicates this with a suggested layout. A table was provided to indicate any flow augmentation and it is hoped that candidates will use such a table wisely in any future question of this type. When using the labelling procedure, the numbers on the network should be lightly crossed through so both the original and adjusted values can be clearly seen by the examiner. The setting out of the solution in part (i) needs much greater attention if candidates are to score full marks for using the labelling procedure in future examinations.

The value of the maximum flow in part (ii) was usually stated correctly but not always shown correctly on the diagram in the insert. Although a few made no attempt at the minimum cut, most were able to find this correctly. Some indicated this cut on a diagram; others indicated that the cut passed through the edges UV, UY, XV, XY ; still others listed the two sets of vertices {S, U, x} and {V, Y, W, Z, T} separated by the cut; each of these methods scored full marks.

Q19.

This question proved to be a good discriminator with the best candidates scoring maximum marks. Weaker candidates often failed to score the marks in parts (a) and (b). Part (a) showed a lack of understanding of cuts. In part (b) candidates should be encouraged to list the paths that they are augmenting at each stage. Although part (c)(i)

was well answered only the best candidates correctly answered part (c)(ii).

Q20.

This was the first time a question had been set with multiple sources. Parts (a) and (b) were, in general, well answered. Nearly all candidates were able to find the value of the cut X in part (c). In part (d)(i) candidates only scored full marks if they showed each separate flow augmentation.

Q21.

Nearly all the candidates were able to find the value of C_1 correctly but there were very few correct answers for C_2 . Part (b) was, in general, well answered. It is helpful for candidates to show each flow augmentation in order to see more easily when an arc has become saturated.

Q22.

This question proved to be far more challenging to candidates than had been expected. Surprisingly, there were very few correct answers to part (a). This showed a lack of understanding as to the value of a cut. Parts (b), (c) and (d) tested candidates understanding of a flow through a network. The numbers were made deliberately easy so as to avoid numerical slips. However, a number of candidates failed to score on these parts. Worryingly, more than one candidate obtained a flow of 40 through the network.